Multi-Agent Hostel Management System

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1. About the Project

This is a hostel management system that uses AI. Students mark attendance and enter the mess using face recognition. The system also looks after mess stock, student complaints and alerts for the warden. Each job is done by a separate AI agent, and all the agents work together.

2.Problem Statement

- Attendance registers and ID cards can be misused, for example one student marking attendance for another.
- Mess staff guess how much food to cook and keep track of stock by hand.
- Complaints are shared in chat groups, so nobody knows which one is urgent and some get missed.
- Attendance, mess and complaint records are kept separately, so the warden finds out late when a student is missing.

3. What We Are Using

PART	TOOLS USED
Backend	Python 3.11 + FastAPI
Database	Supabase (PostgreSQL + pgvector)
Login	Supabase Auth
Face recognition	DeepFace (Facenet512) + OpenCV
Menu PDF reading	Google Gemini
Complaints and stock commands	Groq (Llama 3.3)
Scheduled jobs	APScheduler
Frontend	React + Tailwind CSS
Camera kiosk	Python + OpenCV

4. The Agents

Agent	What it does
IRIS	Registers student faces and recognises them at the gate and mess.
SENTINEL	Records entry and exit, marks students present, late or absent, and makes the list of students who missed attendance.
NOURISH	Lets only registered students into the mess. Reads the menu PDF and keeps track of stock. Staff can update stock by typing, like "added 25 kg rice".
FIXR	Reads a complaint and decides its type (plumbing, electrical, carpentry) and how urgent it is. The warden then assigns a worker.
HERALD	Looks at attendance and meal records together and warns the warden if something looks wrong, like a student missing 3 meals in a row.

5. How It Works

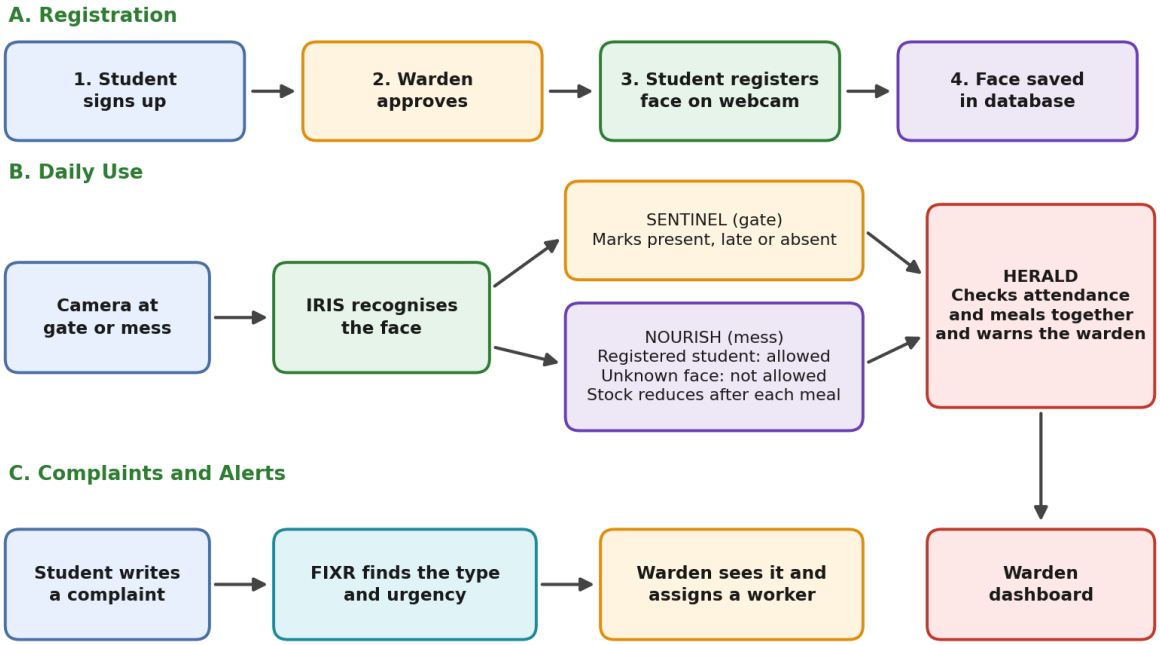


Figure 1: Flow of the project, from student sign up to warden alerts.

- A student signs up, the warden approves, and the student registers their face on a webcam.
- The camera at the gate or mess recognises the face. The gate records attendance and the mess lets the student in.
- Mess staff upload the menu PDF and update stock.
- Students submit complaints, and the warden sees them with the type and urgency.
- After the attendance time is over, the warden gets the list of students who did not mark attendance.

6. What Is Done

- Backend and database, with all five agents working
- Face registration and recognition
- Gate attendance and absent list
- Menu PDF reading, stock management and typed stock commands
- Complaint sorting by type and urgency
- Student, warden and mess staff portals with login

7. What Is Pending

Feature	Details	Status
Live meal count	Use the number of students who entered the mess to calculate food needed.	Next
Stock alerts	Warn mess staff when stock is too low for the menu.	Next
Mark meal as served	One button that reduces stock after a meal.	Next